

# PENDEM VARUN SAI

*ML Systems Engineer | GPU Inference | Computer Vision*

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## EDUCATION

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- **Keshav Memorial Institute of Technology** Hyderabad, India  
*Bachelor of Technology in Computer Science and Engineering* *Aug. 2023 – May 2027 (Expected)*

## EXPERIENCE

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- **Kornia-rs** Remote  
*Open Source Contributor* *March 2026 – July 2026*
  - **KV-Cache Optimization:** Optimized SmolVLM decoding by replacing naive `Tensor::cat` with Candle's `KvCache::append`, cutting 8k-token cache update latency from **78.2 ms to 17.5 ms (4.47× speedup)**.
  - **Attention Acceleration:** Implemented CUDA-accelerated FlashAttention in kornia-rs with automatic CPU/Metal SDPA fallback and mixed-precision (BF16/FP32) handling for the transformer attention layer.
  - **CUDA Compatibility Layer:** Architected a unified CUDA device/dtype dispatch layer for the entire Kornia-ml crate, implementing **BF16 capability detection** with automatic **FP16 fallback** to eliminate unsupported-architecture runtime failures across pre-Ampere NVIDIA GPUs.
- **AOT-Benchmark (NeurIPS '21 & NeurIPS '22 Spotlight)** Remote  
*Sole External Core Contributor* *February 2026 – June 2026*
  - **CPU Fallback & Device Abstraction (8 Merged PRs):** Engineered CPU fallback across CUDA-centric inference pipelines with centralized device routing, device-aware tensor placement, and fault-tolerant checkpoint/data pipelines for cross-platform execution.
  - **Video Tracking & MPS Fallback (4 Approved PRs):** Engineered an end-to-end video tracking pipeline with automated frame extraction, mask validation, and FPS-aware inference while integrating repository-wide Apple Silicon (MPS) fallback support.

## PROJECTS

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- **FalconEye**  
*Multimodal Perception & Real-Time Autonomous Following.*
  - **Prompt-Guided Autonomous Following:** Built a visual tracking system integrating SAM, CLIPSeg, and DaSiamRPN to acquire and track targets via **click, reference image, or text** for autonomous rover following.
  - **High-Throughput TensorRT Inference:** Accelerated DaSiamRPN tracking to **405.8 FPS at 1080p (2.46 ms latency)** and **3,840+ FPS** isolated inference on an RTX 3090 using pinned memory and GPU-native tensors.
  - **Zero-Copy Inference Pipeline:** Eliminated intermediate CPU copies and redundant allocations using **zero-copy ONNX & TensorRT I/O binding**, preallocated device buffers, and explicit CUDA stream synchronization to reduce inference overhead.
- **Segment and Track Anything**  
*Modernized Video Tracking Infrastructure*
  - **Modern Tracking Infrastructure:** Re-engineered **SAM-Track** - the vision backbone adopted by **MIT CSAIL & Harvard SEAS** in FollowAnything (IEEE RA-L) into a modern deployment-grade pipeline
  - **5× Faster Streaming Inference:** Replaced redundant decoding, whole-frame re-segmentation every 3 sec, and per-frame allocator thrashing with a streaming pipeline supporting **mid-stream object insertion** without rollback or tracker restart.
  - **Modern Inference Architecture:** Migrated SAM and Grounding-DINO to **Hugging Face** backends, removing **19K+ lines** of vendored model code while adding lazy loading and CUDA/MPS-aware tracking.

## SKILLS

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- **Languages:** Python, C++, Rust
- **ML / AI:** PyTorch, Transformers, Hugging Face, NumPy, OpenCV, FlashAttention, KV Caching
- **Systems / Runtime:** CUDA, TensorRT, ONNX Runtime, DLPack, Docker, FastAPI, Qt / PySide6, Linux, Git
- **Model Optimization:** Quantization, Automatic Mixed Precision, FP16, BF16, Kernel Optimization